IndicDoc Verify

Deep-Learning Layout Analysis for Pre-Print Checking of Indian Question Papers and Their Translations

Expt.-11: SDG-Based Deep Learning Mini Project

Team

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SDGs addressed SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), SDG 10 (Reduced Inequalities), SDG 16 (Peace, Justice and Strong Institutions)

Code repository `github.com/va-ada/IndicDoc-AI`

Date of report 24 October 2026

Aim

To design, implement, and analyze a Deep Learning solution addressing a real-world problem aligned with the Sustainable Development Goals (SDGs).

Lab Outcome

Students will be able to apply Deep Learning techniques for real world problem-solving and evaluate model performance.

Methodology

The project is carried out in multiple phases to ensure systematic development and implementation:

1. **Topic Selection (SDG Mapping):** Identify a real-world problem aligned with Sustainable Development Goals (SDGs). *Done:* errors in Indian exam papers and their translations, which reach lakhs of candidates (Section 1).
2. **Phase-1: Literature Survey:** Review 4–8 research papers and identify the research gap. *Done:* ten papers from 2021–2026 reviewed, four gaps identified (Sections 2–3).
3. **Phase-2: Final Project Implementation with Performance Analysis:** Develop system architecture, collect dataset, and implement DL model. Analyze results, improve

performance, design front end, integrate the system. Demonstrate the finished project and submit the code along with final working project. *Done:* Sections 4–16.

4. **Documentation:** Prepare a research paper in standard format. *Done:* IEEE conference-format paper in `paper/`.

***Abstract*—Indian exam papers are set in one language and translated into Hindi and regional languages; every version is checked by hand, and errors keep reaching candidates: 49 wrong questions in the Tamil NEET 2018 paper, 59 errors and a re-exam in the 2024 Karnataka civil-services paper, re-tests of three UGC-NET papers in 2026. We present IndicDoc Verify, an offline system that reads a printed paper in any of 12 Indian languages and flags structural and numerical errors, and every difference from its source version, before printing. A deep-learning layout detector finds the questions, sub-questions, options and answers on each page (DocLayout-YOLO, AP@50 0.81–0.90 on these labels, chosen after comparing it with a YOLOv8s and a Faster R-CNN that we fine-tuned); Tesseract reads the text with an English *number pass* that must agree on every digit; the questions are then checked for numbering, duplicates, options, marks and repeats of earlier papers, and two language versions are compared question by question without any translation. On a benchmark of 132 English–Hindi and English–Tamil paper pairs with injected errors, read with real OCR, the system finds 98% of the errors at the right question on born-digital papers (98% on scan-degraded pages) with 3.0 false alarms per clean pair. It ships as a web app, REST API and offline Docker image.

***Index Terms*—document layout analysis, Indian languages, object detection, OCR, question paper verification, translation quality, examinations.

1. Topic Selection and SDG Mapping

Problem. National and state exams in India are bilingual or multilingual: NEET-UG is offered in 13 languages, and UPSC, the state Public Service Commissions, school boards and universities print papers in English and a regional language. Each version is checked by hand before printing. When the check fails, a number, an option, a mark or a whole question differs between versions, and candidates who chose a regional language are judged on a different paper. Table I lists recent cases; each ended in grace marks, a re-exam or a court case. No tool checks a paper and its translations automatically, and a confidential pre-print paper cannot be sent to a cloud service. The mapping to the SDG targets of the 2030 Agenda [20] is given in Table II.

TABLE I
Recent errors in Indian exam papers

Exam	What went wrong	Consequence	Ref.
NEET-UG 2018	Errors in 49 questions of the Tamil version	HC ordered 196 grace marks; Supreme Court set it aside and asked for bilingual subject experts	[21]
NEET-UG 2021	A term dropped in the Hindi version of a physics question	NTA agreed to examine it before the Supreme Court	[22]
KPSC KAS 2024	59 errors in the Kannada paper; machine translation suspected	Re-exam for about 2 lakh candidates, errors again	[23]
APPSC Group-I	51 of 120 questions alleged mistranslated into Telugu	Supreme Court notice	[24]

Exam	What went wrong	Consequence	Ref.
UGC-NET Jun 2026	Factual, typographical and translation errors; questions repeated from earlier papers	Re-test of three papers on 9–10 Sep 2026	[25]
MBBS 2026 (NTRUHS)	The same MCQ printed twice	High Court ordered a compensatory mark	[26]

TABLE II
Mapping of the project to the Sustainable Development Goals

SDG	Target	How IndicDoc Verify contributes
SDG 4 Quality Education	4.3 equal access to tertiary education; 4.5 equal access for the vulnerable	Entrance and board exams in which a regional-language candidate gets the same paper as an English one
SDG 9 Industry, Innovation & Infrastructure	9.5 research and technological capability	Open, offline document-AI for 12 Indian languages, with a measured benchmark
SDG 10 Reduced Inequalities	10.2 inclusion irrespective of ethnicity or origin	Removes a disadvantage that falls only on candidates writing in Indian languages
SDG 16 Strong Institutions	16.6 effective, accountable institutions; 16.10 access to information	Exam bodies catch errors before they become litigation; bilingual circulars checked for compliance

2. Literature Survey Summary Table

Ten papers from 2021–2026 were reviewed (Table III): Indian-language document layout and recognition [1]–[8], from Phase-1, and two works on checking translations [9], [10], added when the project turned to exam papers.

TABLE III
Literature survey summary

Ref	Author	Problem	Dataset	Method	Limitation
[1]	Nath *et al.*, 2025	Layout parsing of Indian-language pages	IndicDLP: 119,806 pages, 12 languages, 42 labels incl. questions and options	YOLOv10x, DocLayout-YOLO, RoDLA, DINO, DiT, Florence-2	Regions only; question labels not used to check papers
[2]	Kasuba *et al.*, 2025	Page-level handwritten Indic text recognition	CHIPS, ~25K pages, 10 languages	DBNet word detection + 6 recognisers	User must name the language; text only
[3]	Mondal *et al.*, 2026	Multilingual table detection	UniTabBank: 55K pages, 28 languages	Semi-automatic labelling; YOLOv11	Borderless tables and degraded scans fail
[4]	Pal *et al.*, 2024	Table QA for low-resource Indic languages	BanglaTabQA 2M, HindiTabQA 643K	Fine-tuned mBART / M2M / Llama	Wikipedia tables only
[5]	Xu *et al.*, 2021	Multilingual document understanding	30M documents; XFUND, 7 languages	LayoutXLM: text + layout + image	No Indian language
[6]	Kasuba *et al.*, 2025	Grounding answers on the page	MGVG: 70 circulars, 509 QA pairs	OCR + Llama-3.1-8B + region matching	OCR-dependent; single page
[7]	Bhattacharyya *et al.*, 2025	Hindi OCR with vision-language models	1.1M printed Hindi words	LoRA-tuned GOT-OCR 2.0	Hindi only; mixed-language lines hard

Ref	Author	Problem	Dataset	Method	Limitation
[8]	Dumada & Shah, 2026	Devanagari table extraction	500+ documents	Structure + Tesseract, weighted fusion	No ablation
[9]	Babayeva *et al.*, 2025	Translating a test into other languages with an LLM	Force Concept Inventory	GPT-4o + back-translation + human review	Every item still checked by hand; no page structure
[10]	Rei *et al.*, 2022	Reference-free translation quality estimation	WMT 2022 QE task	CometKiwi: multilingual encoder scores sentences	Plain sentences: no numbers, options or marks

3. Research Gaps

1. **G1: No automatic pre-print check.** Papers are translated into up to 13 languages and vetted by hand; the cases in Table I show that this step fails.
2. **G2: Translation tools ignore the page.** Quality estimation [10] and LLM back-translation [9] work on plain sentences; they cannot see question numbers, options, marks or figures, they need clean text, and they run in the cloud.
3. **G3: Layout models are not used for checking.** IndicDLP [1] labels questions, options and answers in 12 languages, but detected structure has not been used to verify a document.
4. **G4: Numbers are unreliable in Indic OCR.** Indic models misread Latin digits in mixed text [7]; a checker of numbers needs every digit right.

Choice of problem. We also examined court-filing scrutiny, where AI defect detection is already deployed (the Supreme Court with IIT Madras; a commercial tool is reported in 16 High Courts [27]), and chose instead the problem that no tool addresses yet.

4. Problem Statement and Objectives

Problem statement. Given the pages of a question paper in any of 12 Indian languages, and optionally its source-language version, find every structural or numerical error and every difference between the versions before the paper is printed, and show where each one is.

Main objective. Build an offline deep-learning system that verifies a question paper and its translations, and measure how many real kinds of error it finds and how many false alarms it raises.

Specific objectives.

1. Detect question-paper structure with a layout detector; fine-tune two detectors and compare them with the released IndicDLP model.
2. Read text in 12 languages with reliable numbers (number pass).
3. Implement structure, translation-parity, repeat and format checks with a report that points to the page.
4. Evaluate on a benchmark with injected errors; deliver a web app, REST API and Docker image.

TABLE IV
Research questions

RQ	Question
RQ1	How well do deep-learning layout detectors recover the structure of Indian question papers?
RQ2	What share of real kinds of error does layout-aware checking find, at what false-alarm rate?
RQ3	What do the number pass and layout-based option splitting contribute?
RQ4	How robust is it to scanned pages and to layout-model mistakes?

Expected output. A verdict (ready / check / do not print), each finding with its question and page, the pages with every problem outlined, as HTML and JSON.

5. Proposed System, Methodology and Novelty

Workflow (input to output).

1. **Input:** the paper as PDF (rendered at 200 dpi) or scans; optionally its source version, a question bank of earlier papers and a format profile.
2. **Find the structure:** DocLayout-YOLO [11] detects questions, sub-questions, options, answers, figures, tables and page furniture on every page (high-recall threshold chosen on validation pages).
3. **Read the text:** Tesseract [19] reads the page with its own script's model; an English *number pass* re-reads it and fixes the Latin digits the Indic model misread. Words go to the region that contains them.
4. **Build the questions:** a question region starts a question; options are separated by the gaps between words (layout), so garbled option markers do not matter; marks, figures, tables and formulas are attached.
5. **Check:** numbering, duplicates, options, marks total; repeats of earlier questions; parity with the source version; format rules.
6. **Report:** verdict and findings, each outlined on its page.

How deep learning is used. The structure of a page is found by a convolutional object detector fine-tuned on Indian-language pages (transfer learning from COCO and synthetic documents). Its regions, and the word positions inside them, turn a picture of a page in any script into structured questions; the checks then work on that structure.

Novelty. (i) *Translation parity without translation:* numbers, option counts, marks, sub-questions and figures are the same in every language, so two versions are aligned question by question (Needleman–Wunsch on these language-independent features) and compared without a translation model. (ii) *Layout-aware checking:* detected question and option regions plus word geometry, instead of plain text. (iii) *Number pass:* two OCR models must agree on every digit, with page-level detection of whether the paper prints Latin or Indian digits. (iv) *Offline and general:* one Docker image, 12 languages, and JSON format profiles for other documents (e.g. bilingual government circulars).

6. System Block Diagram

Fig. 1 shows the components and the flow of data. The app, the API and the benchmark run the same code.

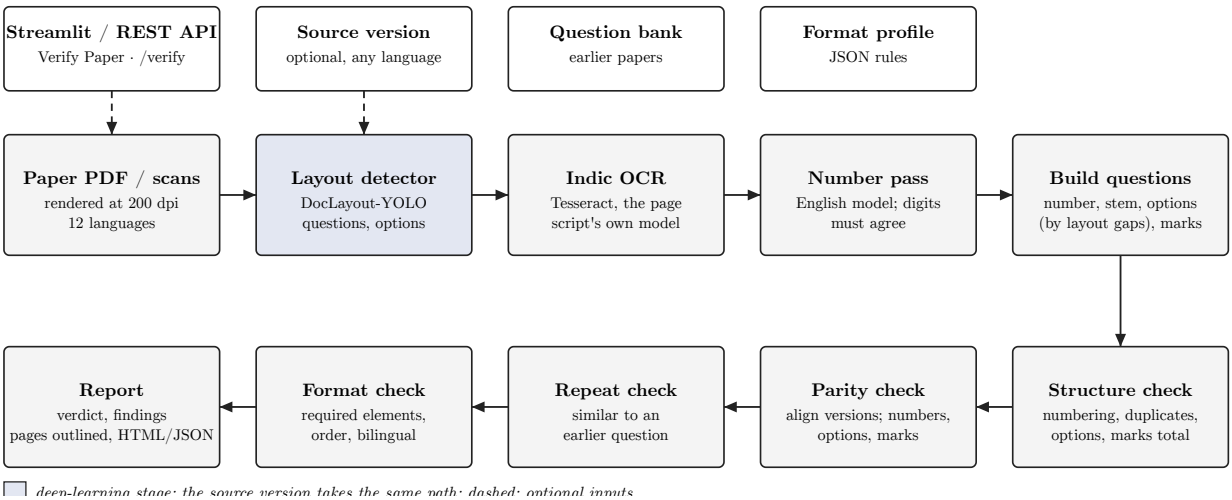


Fig. 1. System block diagram of IndicDoc Verify. The paper flows left to right through the deep-learning layout stage and OCR, then right to left through the checks to the report. The question bank feeds the repeat check and the profile the format check.

7. Dataset Description and Pre-processing

Layout data. IndicDLP [1], [17]: 119,806 human-annotated pages, 12 languages, 12 domains, 42 labels, including first-, second- and third-level questions, options and answers. Our language-balanced subset has 3,317 pages and 49,995 boxes from the *official* splits (Table V); 195 pages are question papers, with 1,227 question and 1,209 option boxes. It is released as CSV files in `dataset/`.

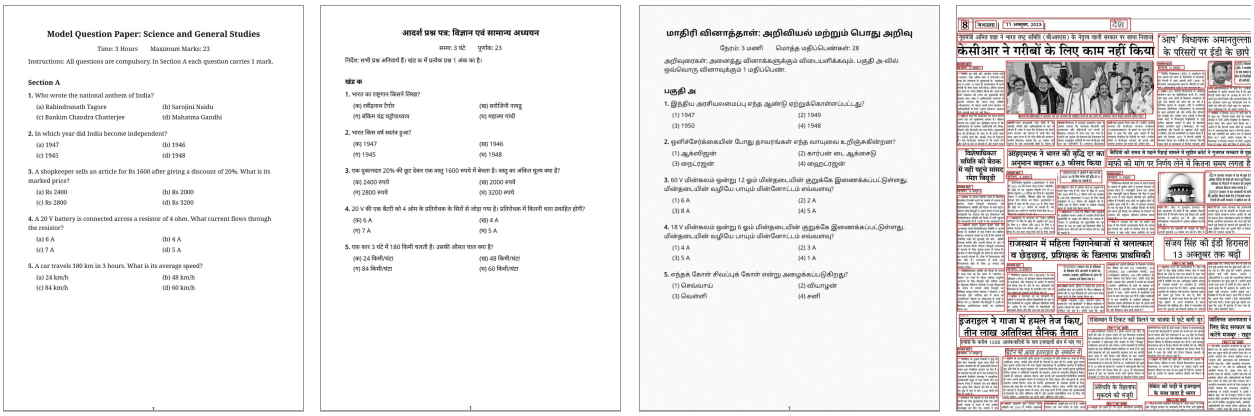
TABLE V
INDICDLP SUBSET USED FOR THE LAYOUT MODELS (PAGES PER SPLIT)

Language	**Train**	**Validation**	**Test**	**Total**
Assamese	129	30	60	219
Bengali	200	30	60	290
English	188	30	60	278
Gujarati	200	30	60	290
Hindi	200	21	22	243
Kannada	200	21	60	281
Malayalam	200	30	60	290
Marathi	200	30	60	290
Odia	200	30	60	290
Punjabi	188	30	60	278
Tamil	200	30	60	290
Telugu	188	30	60	278
Total pages	**2,293**	**342**	**682**	**3,317**
Labelled boxes	33,667	5,815	10,513	49,995

Verification benchmark. No exam body publishes erroneous pre-print papers, so we built one. Each of the 132 pairs is an English source and its Hindi or Tamil version: 18 questions (15 MCQs from numeric templates and fixed questions, 3 short-answer questions with marks), set like a state-board model paper and rendered at 200 dpi with Noto fonts. 108 pairs carry one injected error of the kinds in Table I (Table VI); 24 are clean, to count false alarms. Half of the pairs are degraded like scans (Gaussian blur 0.7 px, noise $\sigma = 8$, JPEG quality 55). A separate *development* set (48 pairs, other random seed) was used to build the parser; the test set was run once. Fig. 2 shows sample pages.

TABLE VI
INJECTED ERRORS IN THE VERIFICATION BENCHMARK (TEST SET)

Error	**How it is made**	**Pairs**
Number changed in stem	one value in a translated stem changed	12
Number changed in an option	one option value changed in the translation	12
Option missing	one option removed from the translation	12
Question missing	a question removed (numbers renumbered or left with a gap)	12
Question printed twice	an MCQ printed again in place of another (both versions)	12
Marks differ	marks of a short-answer question changed in the translation	12
Wrong maximum marks	maximum marks changed in the translation	12
Question numbering error	a question printed with the previous number	12
Repeated from a past paper	a question copied from a "previous year" paper (both versions)	12



English source *Hindi version* *Tamil version, scan-degraded* *IndicDLP page (Hindi)*

Fig. 2. Sample pages: the verification benchmark (left three) and IndicDLP.

TABLE VII
PRE-PROCESSING AND AUGMENTATION

Step	**What was done**
Layout data	Parquet row groups streamed from Hugging Face; PNG + COCO JSON; boxes clipped, <1 px boxes dropped; COCO → YOLO labels
Resizing	Letterbox to 640 px (our models) or 1024 px (DocLayout-YOLO)
Augmentation	YOLOv8s: mosaic, HSV, scale, translate; Faster R-CNN: colour jitter, blur; no mirror flips
Verification input	PDF pages rendered at 200 dpi so every digit is legible; images used as they are
Scan degradation	Gaussian blur 0.7 px, Gaussian noise $\sigma = 8$, JPEG quality 55, on half of the benchmark

Challenges. Class imbalance (*paragraph* is 32% of boxes); no OCR ground truth in IndicDLP; no public data of real erroneous papers, hence the benchmark; Indic OCR of Latin digits (Section 15).

8. Deep Learning Model Architecture

Three detectors were compared (Fig. 3, Table VIII). All follow the backbone–neck–head design. The released DocLayout-YOLO is used by IndicDoc Verify; its AP@50 on the question-paper labels is in Table XIII.

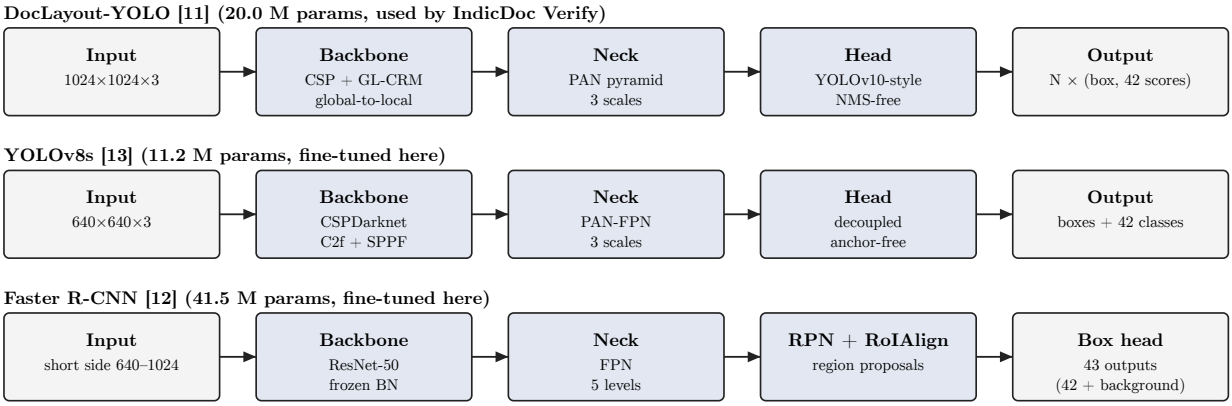


Fig. 3. Architectures of the three detectors with input and output dimensions. Shaded blocks are the learned feature extractor and prediction layers.

TABLE VIII
MODEL DETAILS

	DocLayout-YOLO	**YOLOv8s**	**Faster R-CNN**
Role	used by IndicDoc Verify	fine-tuned (proposed)	fine-tuned (baseline)
Type	one-stage, NMS-free	one-stage, anchor-free	two-stage
Parameters	20.0 M	11.2 M	41.5 M
Input	1024 × 1024 × 3	640 × 640 × 3	short side 640–1024
Output	N × (box, 42 scores)	N × (box, 42 scores)	N × (box, 43 scores)
Pre-training	synthetic documents, then full IndicDLP [18]	COCO	COCO
Modification	none	80-class head → 42 classes	predictor → 43 classes
Loss	—	CIoU + DFL + BCE	RPN + cross-entropy + smooth-L1

9. Tools, Technologies and Environment

TABLE IX
TOOLS, TECHNOLOGIES AND ENVIRONMENT

Component	**Details**
Programming language	Python 3.11 (models, checks, app, API); JavaScript (Node) for the benchmark renderer
DL framework	PyTorch 2 [15] with torchvision; Ultralytics 8 (YOLOv8) [13]; doclayout-yolo 0.0.4
Development environment	VS Code and terminal, Git + GitHub, pytest (unit tests for every check)
Libraries	pycocotools, OpenCV, Pillow, NumPy, pandas, Matplotlib, PyMuPDF; Tesseract 5 (tessdata_best, 12 scripts) [19]
Hardware	Apple M1 Pro GPU (MPS) for training and layout evaluation; CPU container for the verification benchmark
Dataset tools	Hugging Face Hub + fsspec; COCO / YOLO converters; Playwright and Noto fonts to render benchmark papers; tesseract.js

Component	Details
Interface / deployment	Streamlit app (Verify Paper tab), FastAPI (POST /verify), Docker + docker-compose, offline

10. Implementation Details

TABLE X
IMPLEMENTATION SUMMARY

Part	Implementation (file)
Layout training	COCO-pretrained YOLOv8s (batch 8, 12 epochs, 60 min) and Faster R-CNN (SGD lr 0.005, momentum 0.9, weight decay 5×10^{-4}, cosine, batch 4, 2 epochs, 76 min); best checkpoint on 342 validation pages (`ml/training/`)
Inference	DocLayout-YOLO at its high-recall threshold, duplicate boxes merged (`ml/inference/predict.py`)
OCR + number pass	Indic model + English re-read; digits replaced only where the Indic word has no Indic letters or the English reading is more confident; page-level Latin / Indian digit decision (`ml/ocr/`)
Question parsing	question regions, layout-gap option splitting, marker stripping, marks, figures; merged numbers ("5.16" $\rightarrow$ 5, 16) split by sequence (`ml/verify/paper.py`)
Checks	structure, parity (global alignment), repeats (character 3-gram cosine), format profiles (`ml/verify/checks.py, profiles.py`)
Report and interface	HTML / JSON report with outlined pages; Streamlit tab; `POST /verify`
Benchmark	`scripts/verify_bench/` (generate, render, degrade, OCR), `scripts/evaluate_verify.py`

Listing 1. Verifying a Hindi paper against its English source.

```
# every page: layout model, then OCR with the number pass
out = predict_single(model, page, "doclayout", device, conf=high_recall)
doc = extract_document(page, out, "hin", number_pass=True)

# ml/verify: questions, checks, parity, repeats, report
report = verify(hindi_pages, source_pages=english_pages, bank=earlier_questions)
print(report["summary"])            # {'questions': 18, 'errors': 1, 'warnings': 0, ...}
html = to_html(report, {"paper": page_images})
```

11. Experimental Setup and Training

TABLE XI
EXPERIMENTAL SETUP

Experiment	Setting
E1 Layout	3 detectors on 682 held-out IndicDLP pages (10,513 boxes): COCO mAP, P / R / F1, per label
E2 Robustness of layout	blur, noise, JPEG, half resolution, faded print (as in RoDLA [14]); Robustness Index = damaged / clean mAP@50
E3 Verification	132 paper pairs; layout boxes from the renderer, real Tesseract OCR; an error is found only if a finding of the matching kind points at the right question (±1) or, for paper-level errors, at the paper
E4 Ablations	without the number pass; without layout-based option splitting; noisy layout (boxes shifted by up to 8 px, 10% of question / option labels changed to "paragraph")

Experiment	Setting
Protocol	parser developed on a separate development set; test set run once

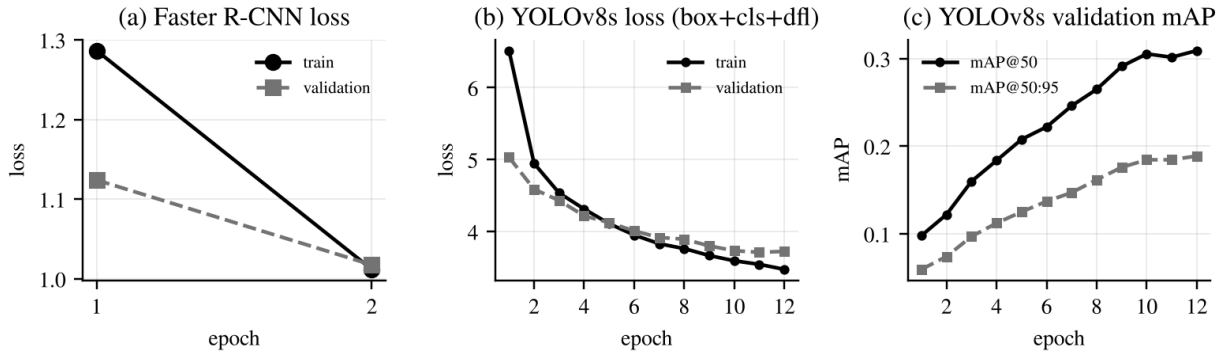


Fig. 4. Training curves of our two detectors (measured): (a) Faster R-CNN loss; (b) YOLOv8s loss; (c) YOLOv8s validation mAP per epoch.

12. Results and Performance Evaluation

RQ1: structure. Table XII gives the layout results on the same 682 test pages; Table XIII the question-paper labels. The released model finds questions and options reliably (AP@50 0.81–0.90; question-paper domain mAP@50 0.84), which is why the checker is built on it.

TABLE XII
Layout results on 682 test pages

Metric	DocLayout-YOLO	YOLOv8s (ours)	Faster R-CNN (ours)
mAP@50	**0.759**	0.259	0.148
mAP@50:95	**0.575**	0.165	0.079
Precision (balanced threshold)	**0.848**	0.603	0.510
Recall (balanced threshold)	**0.775**	0.508	0.435
F1 (balanced threshold)	**0.810**	0.551	0.469
Recall, high-recall threshold	**0.851**	0.508	0.435
Inference time (s / page)	0.150	**0.024**	0.202

TABLE XIII
AP@50 on the question-paper labels

Label	DocLayout-YOLO	YOLOv8s	Faster R-CNN
first-level-question	0.854	0.415	0.240
second-level-question	0.898	0.045	0.018
third-level-question	0.872	0.026	0.000
options	0.839	0.553	0.243
answer	0.809	0.000	0.000

RQ2: verification. Table XIV summarises the test set. On born-digital papers (the normal pre-print case) 98% of the 59 injected errors are found at the right question, with 3.0 false alarms

per clean pair; on scan-degraded pages 98% with 5.5. Fig. 5 breaks the result down by error type. The questions were recovered from the pages in 100% of cases. The only misses were two Hindi papers whose "maximum marks" line OCR read as "पूर्णांक" (anusvara dropped). After the test run we widened that pattern; re-running the unchanged test set then finds 100% with 4.38 false alarms per clean pair. We report the frozen numbers above and this post-hoc result separately.

TABLE XIV
VERIFICATION RESULTS ON THE TEST SET (132 PAPER PAIRS)

Condition	Errors	Found at right question	Clean pairs	False alarms / clean pair
Born-digital (clean render)	59	98%	10	3.00
Scan-degraded	49	98%	14	5.50
English → Hindi	54	96%	12	7.00
English → Tamil	54	100%	12	1.92
All	108	**98%**	24	**4.46**

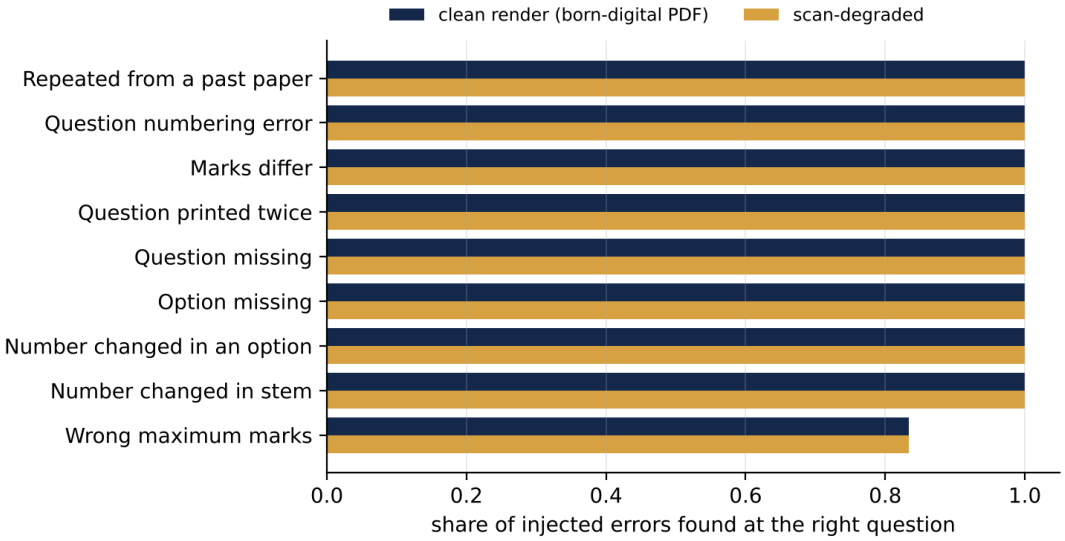


Fig. 5. Share of injected errors found at the right question, per error type, on born-digital and scan-degraded papers (test set).

RQ3 and RQ4: ablations and robustness. Without the number pass, false alarms rise from 4.5 to 13.8 per clean pair (Table XV, Fig. 6): Indic OCR of Latin digits is the main source of noise. With a noisy layout, 98% of errors are still found, because the parser recovers questions from their text when a label is wrong. The layout model itself keeps 97% of its accuracy on damaged scans (E2).

TABLE XV
Ablations on the test set

System	Found at right question	False alarms / clean pair
Full system	98%	4.46
Without number pass	77%	13.75
Without layout-based options	98%	11.46
Noisy layout	98%	7.50

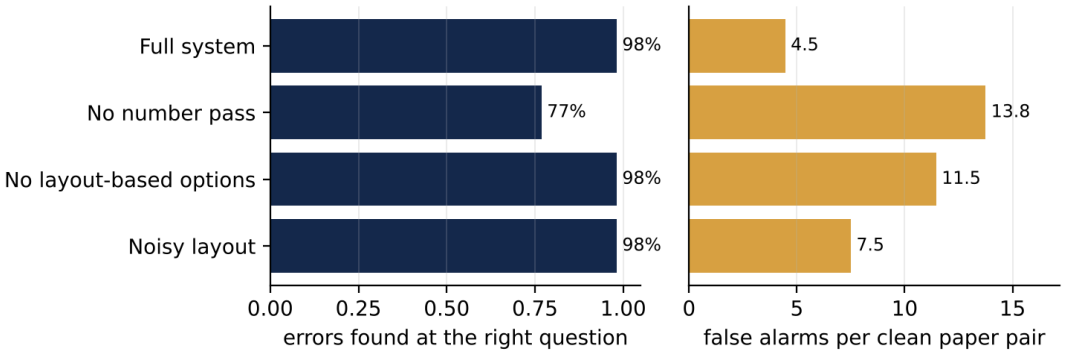


Fig. 6. Ablations: errors found (left) and false alarms (right).

13. Comparison with Baseline / Existing Methods

TABLE XVI
Layout detectors and published results

Method	Training data	mAP@50	mAP@50:95
Faster R-CNN (ours, baseline)	2,293 pages, 76 min	0.148	0.079
YOLOv8s (ours, proposed)	2,293 pages, 60 min	0.259	0.165
DocLayout-YOLO (released by [1])	~96K pages	**0.759**	**0.575**
Published: DocLayout-YOLO [1]	full IndicDLP test set	≈ 0.735	0.545

With the same data and a similar budget, YOLOv8s beats the Faster R-CNN baseline on every metric and is 9× faster; the released model, trained on about 42× more pages, is best and agrees with the published result, so IndicDoc Verify uses it. Table XVII compares the system with existing ways of checking papers; none of them compares language versions of a paper question by question, offline.

TABLE XVII
INDICDOC VERIFY AND EXISTING APPROACHES

Approach	Checks the paper	Compares versions	Indic, offline
Manual vetting (status quo)	yes, by hand	by hand	yes
Answer-sheet evaluation tools	no (grades answers)	no	partly
MT quality estimation [10]	no structure	sentence by sentence	GPU / cloud
LLM back-translation [9]	no structure	with human review	cloud
IndicDoc Verify	**numbering, options, marks, repeats**	**question by question**	**12 languages, offline**

14. Sample Outputs and Demonstration

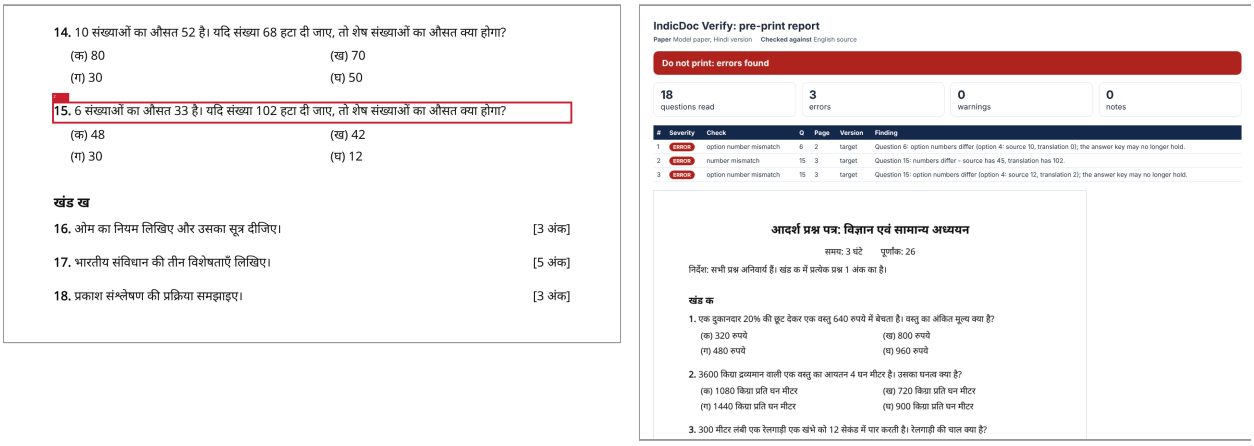


Fig. 7. Correct output. Left: a number changed in the Hindi version, found and outlined on the page. Right: the report with the verdict and the findings.

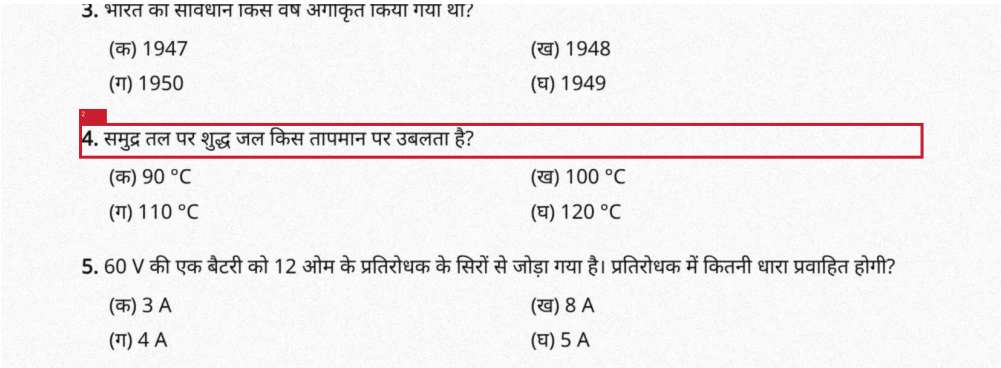


Fig. 8. Incorrect output: a false alarm on a scan-degraded page, where OCR lost part of an option value so the versions appear to differ.

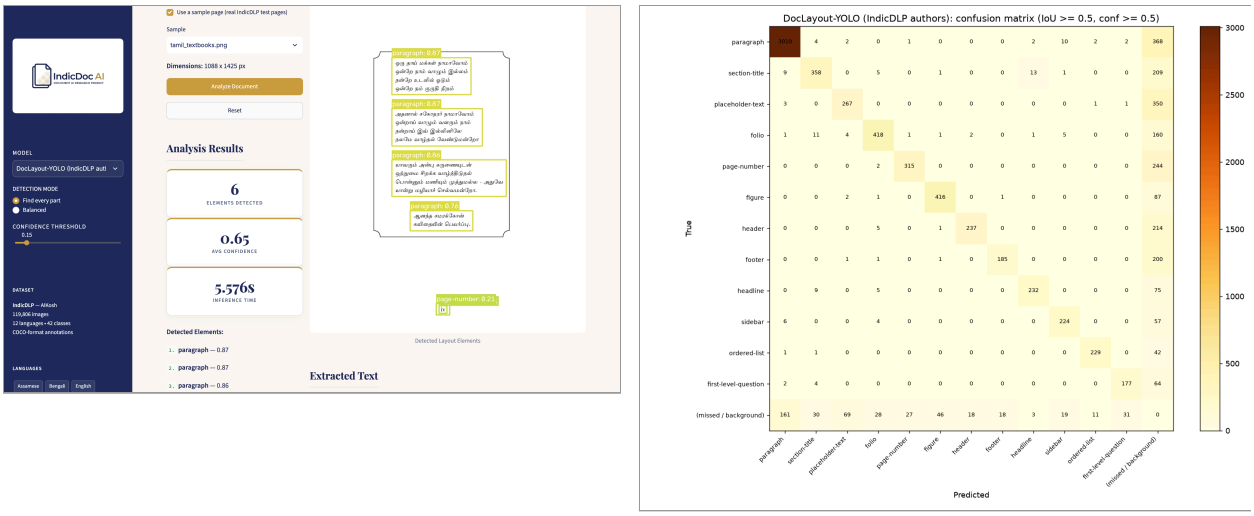


Fig. 9. Layout model in the running app (Tamil page) and its confusion matrix: most errors are missed regions.

15. Challenges and Limitations

1. **OCR of numbers.** The Hindi model reads "1947" as "947", "1" as "।" and units as Indian digits; markers run into values. The number pass, layout-based options and the page-level digit decision were needed; scan-degraded pages still give more false alarms.
2. **No real error data.** Erroneous pre-print papers are not public; the benchmark injects realistic errors into rendered papers, with oracle layout boxes (the noisy-layout run tests the layout dependence).
3. **Meaning.** Changes of meaning without a change of number or structure need a cross-lingual sentence model; the hook exists but is not evaluated. Number words and formulas are not compared yet.
4. **Compute.** Our own detectors trained on 2,293 pages for about an hour and stay below the released model; the product uses the released model.

16. Conclusion and Future Scope

Conclusion. IndicDoc Verify checks an Indian question paper and its translations before printing. A deep-learning layout detector recovers the questions and options (AP@50 0.81–0.90); OCR with a number pass reads them; structure, parity, repeat and format checks find 98% of injected errors at the right question on born-digital papers (98% on degraded scans) with 3.0 false alarms per clean pair. All four objectives were met; our own detectors did not reach the released model, which we report openly.

Product. Buyers are national exam agencies, UPSC and state commissions, school boards, university exam cells and test-series publishers. Because pre-print papers are confidential, the product is an offline licence per exam body; a hosted plan suits only non-confidential mock tests.

Future scope.

- Pilot with a university exam cell or a state board on real pre-print papers; measure minutes saved.
- A cross-lingual meaning check (sentence embeddings or quality estimation [10]) as a warning for the vetter.
- Better Indic OCR behind the same interface; number words, formulas and answer-key consistency.
- More format profiles: bilingual circulars, forms, certificates; a question bank built from MILU-style data [16].

Expected Outcomes

TABLE XVIII
EXPECTED OUTCOMES AND THEIR STATUS

Expected outcome	**Status**	**Evidence**
Development of a working Deep Learning model addressing a real-world problem	Achieved	Layout detectors + IndicDoc Verify in a working app and API (Sections 8–10, 14)
Performance evaluation of implemented model	Achieved	Layout mAP and per-label AP; errors found and false alarms on a benchmark; ablations (Sections 12–13)
Improved understanding of research methodology and problem-solving skills	Achieved	Research questions, separate development and test sets, honest failure cases
Preparation of a research paper suitable for submission	Achieved	IEEE-format paper in `paper/`

Deliverables

TABLE XIX
DELIVERABLES AND THEIR LOCATION IN THE REPOSITORY

Deliverable	**Location**	**Contents**
Dataset (CSV / image dataset)	`dataset/`, `scripts/verify_bench/`	IndicDLP subset as CSV (3,317 pages, 49,995 boxes); generator of the verification benchmark
Project code with complete implementation	`ml/ app/ backend/ scripts/`	Training, evaluation, OCR, verification, web app, REST API, Docker [28]
GitHub repository with organized code and documentation	`README.md, docs/`	Product, dataset, architecture, evaluation and results documents; tests
Presentation slides (Phase-1 and Phase-2)	`docs/presentation/`	PDF and PPTX
Final research paper	`paper/`	IEEE-format paper
Experiment report (this document)	`docs/report/`	generated by `scripts/build_lab_report.py`

Evaluation Criteria

TABLE XX
EVALUATION CRITERIA AND WHERE EACH IS ADDRESSED

Criterion	Marks	Where addressed
Relevance of problem to SDG	2	Section 1, Tables I–II (SDG 4, 9, 10, 16)
Quality of literature survey and research gap identification	2	Sections 2–3
Implementation of deep learning models	2	Sections 7–11
Performance evaluation and Frontend development	2	Sections 12–14; Streamlit Verify tab, REST API
Documentation and presentation quality	2	Repository docs, Phase-1 / Phase-2 slides, this report
Innovation and Research Paper	5	Translation parity without translation, number pass, layout-based checking; IEEE paper

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